



GS FOUNDATION 2024-25

GEOGRAPHY - 23

Physical Setting of India - I

TABLE OF CONTENTS

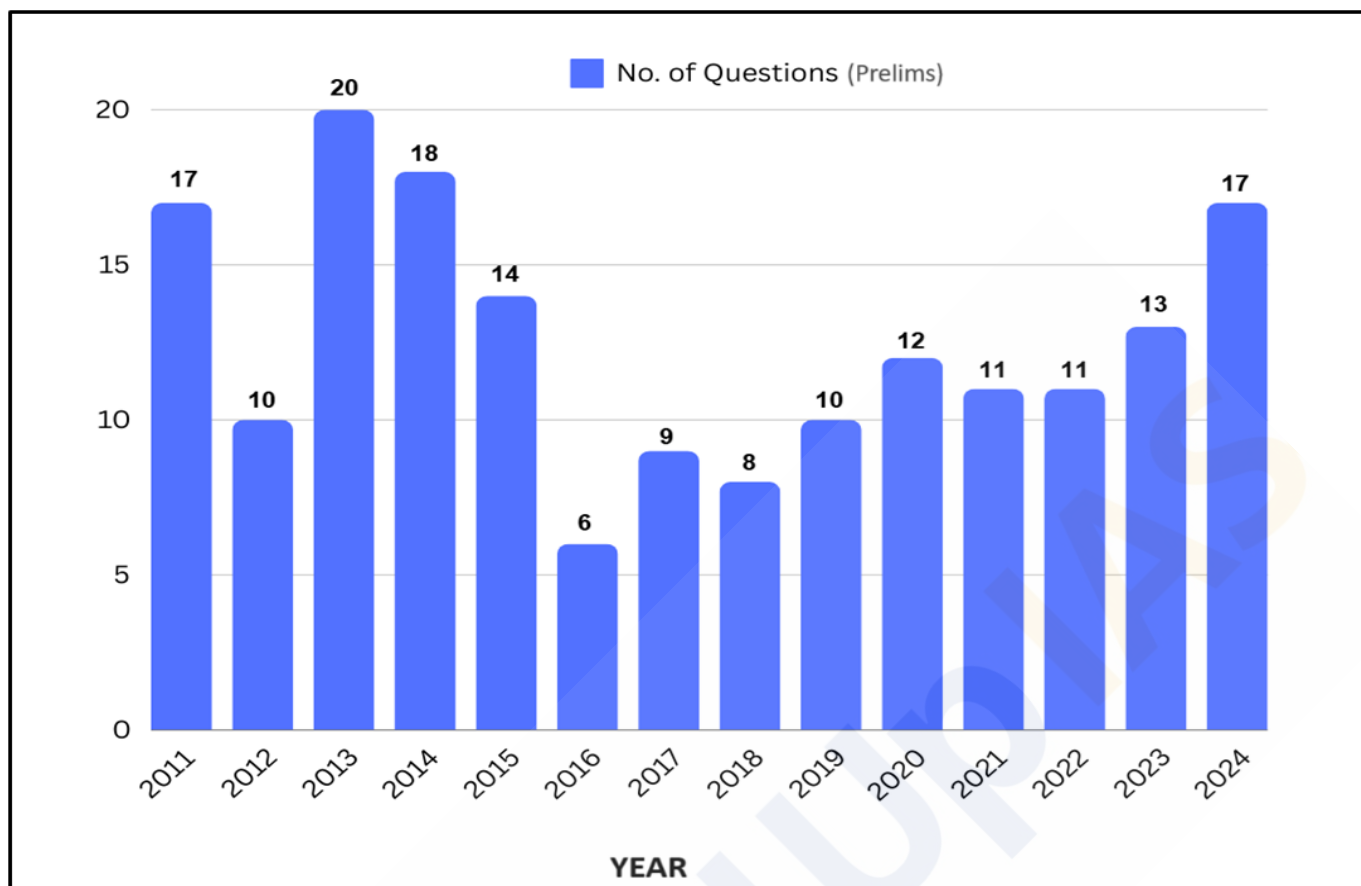
Size and Location

Introduction	3
Nomenclature of India	3
Location of India	3
<i>Tropical and Subtropical Zones</i>	<i>4</i>
<i>Peninsular Shape and Maritime Influence</i>	<i>4</i>
<i>Time Zone and Standard Time</i>	<i>5</i>
Size of India	7
India and its Neighbours	8
<i>Border with China</i>	<i>8</i>
<i>Border with Nepal</i>	<i>9</i>
<i>Border with Pakistan</i>	<i>9</i>
<i>Border with Bangladesh</i>	<i>10</i>
<i>Border with Myanmar</i>	<i>10</i>
<i>Border with Sri Lanka</i>	<i>10</i>

Geological Structure

Introduction	11
Geological Structure and Time Scale	11
Major Rock Systems in India	11
<i>Archaean Rock System (Pre-Cambrian Rocks)</i>	<i>11</i>
<i>Purana Rock System</i>	<i>12</i>
<i>Dravidian Rock System (Palaeozoic)</i>	<i>12</i>
<i>Aryan Rock System</i>	<i>12</i>

Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

Physical Geography

Topic	Questions
Geomorphology	9
Oceanography	5
Climatology	22
Mapping	16
Distribution of Resources	5
Miscellaneous	15

Indian Geography

Topic	Questions
Physiography of India	24
Drainage System of India	15
Indian Climate	7
Indian Soils	5
Natural Vegetation of India	12
Indian Agriculture	31
Mineral Resources of India	10

Size and Location

Introduction:

India is a country of immense geographical diversity, stretching from the icy **peaks of the Himalayas in the north** to the **tropical shores of the Indian Ocean in the south**. Located in Asia, the largest continent in the world, India forms a significant part of South Asia. It is geographically separated from the rest of the continent by the towering Himalayas, which contribute to the country's unique climatic and topographical features. India's vast landmass is home to a **variety of landscapes, ranging from the snow-covered heights of the northern Himalayas to the fertile plains of the Indo-Gangetic region, and the arid expanses of the Thar Desert in the west**. The southern part of the country is dominated by the Peninsular Plateau, which is flanked by the Eastern and Western Coastal Plains. The tropical sun and monsoonal rains play a crucial role in shaping the climate and life in India, making these two climatic factors vital for the country's people. Due to its immense size and diversity, **India is often referred to as a subcontinent**, possessing many characteristics typically associated with a continent.

Nomenclature of India:

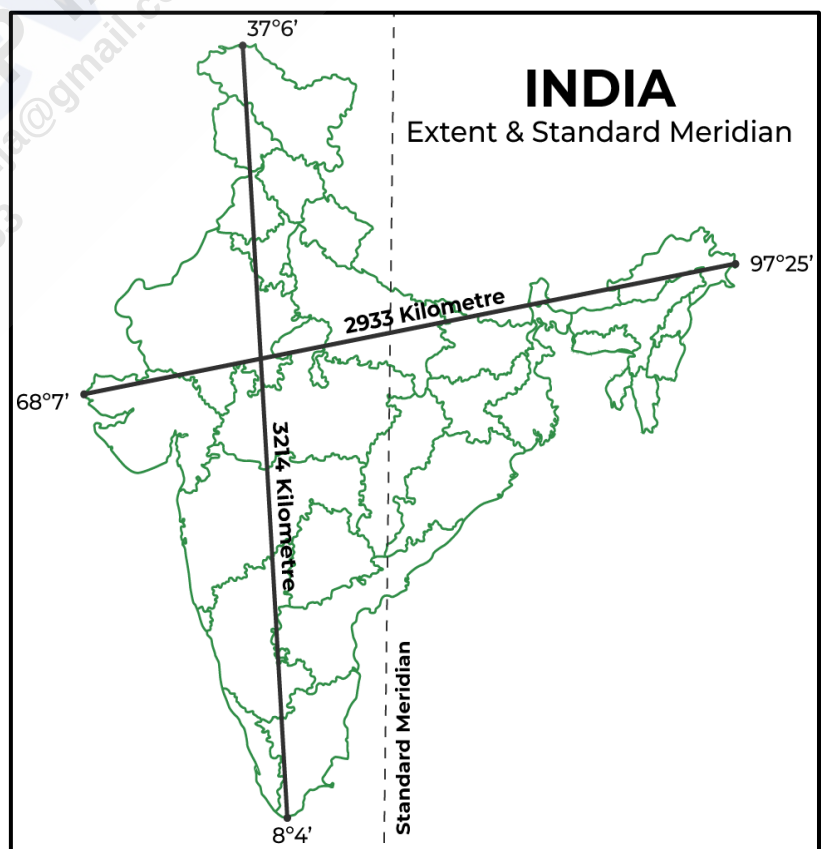
The country has been known by various names throughout history, including **India, Hindustan, Bharat, and Aryavarta**. The name "**India**" originates from Greek literature, referring to the land of the "**Indoi**," or **the people living near the Indus River**. The Persians and Greeks extended the name "**Sindhu**," the river, to the entire region, which came to be known as **Hindustan** in Persian and other West Asian languages. The term "Hindu" is derived from "**Sindhu**," with the Persians pronouncing "S" as "H," leading to the name "**Hindu**" for the land to the east of the Sindhu River.

In Hindu literature, the subcontinent is referred to as **Bharat or Bharat-Varsha**, named after the legendary **King Bharata** who is said to have visualized the fundamental unity of the country. Some scholars believe the name "**Bharat**" is derived from the **Bharath tribe** that inhabited the region. In European languages, the country is popularly known as India, while "**Aryavarta**" refers to the land of the Aryan race. Today, both "**India**" and "**Bharat**" are officially recognized names of the country, although "**Hindustan**" remains in common use.

Location of India:

India is **entirely located in the Northern Hemisphere**, with the Indian mainland **extending from 8°4'N to 37°6'N latitudes** and from **68°7'E to 97°25'E longitudes**. This results in a latitudinal and longitudinal extent of about **29 degrees**.

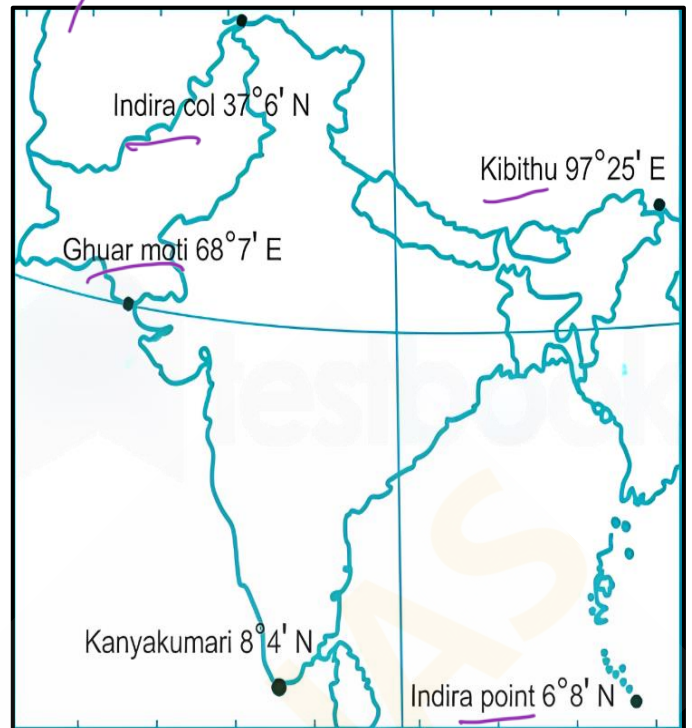
The country measures **approximately 3,214 kilometers from north to south** and **2,933 kilometers from east to west**. However, despite the similar extent in degrees, the actual distances differ considerably due to the **curvature of the Earth**. The east-west distance between two successive meridians of longitude along the equator is at its maximum—about 111 kilometers—and gradually decreases as one



moves toward the poles, where the distance becomes zero. This is because all meridians converge at the poles. In contrast, the north-south distance between any two successive parallels of latitude along a meridian remains relatively uniform at **approximately 111 kilometers**.

The **northernmost point of the Indian mainland** is located in **Jammu and Kashmir (Indira Col)**, while the **southernmost point is Kanyakumari in Tamil Nadu**. However, the southernmost point of the entire country is **Indira Point** in the Andaman and Nicobar Islands, situated at **6°30'N latitude**.

The **westernmost point of India** is in **Gujarat (Ghuar Moti)**, and the **easternmost point is in Arunachal Pradesh (Kibithu)**.



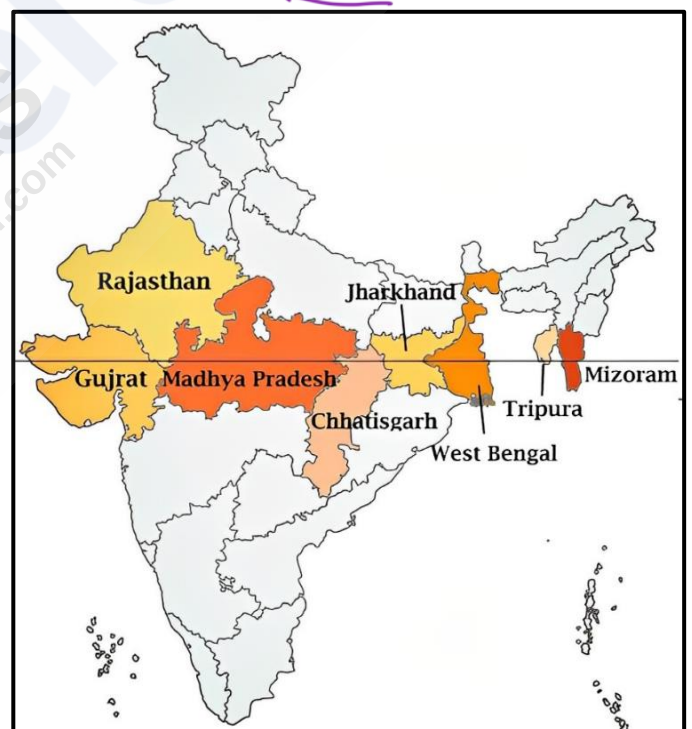
India's large latitudinal extent significantly affects the climate and the length of day and night across the country. The northern parts of India, being farther from the equator, receive the Sun's rays more obliquely, resulting in less insolation and a colder climate compared to the southern parts.

Additionally, the **difference between the length of day and night** varies considerably across the country. In the **southernmost parts of India, near the equator, the difference is minimal—about 45 minutes—**while in the **northern parts, it can be as much as 5 hours**.

Tropical and Subtropical Zones:

The **Tropic of Cancer** passes almost midway through the country, dividing it into **two climatic zones**. The southern half of India, lying south of the Tropic of Cancer, is situated in the **Tropical or Torrid Zone**, characterized by a warm climate. The northern half, lying north of the Tropic of Cancer, falls in the **Subtropical Zone**, which experiences a relatively cooler climate.

The Tropic of Cancer passes through **8 states in India** - **Gujarat** (Jasdan), **Rajasthan** (Kalinjarh), **Madhya Pradesh** (Shajapur), **Chhattisgarh** (Sonhat), **Jharkhand** (Lohardaga), **West-Bengal** (Krishnanagar), **Tripura** (Udaipur) and **Mizoram** (Champhai).



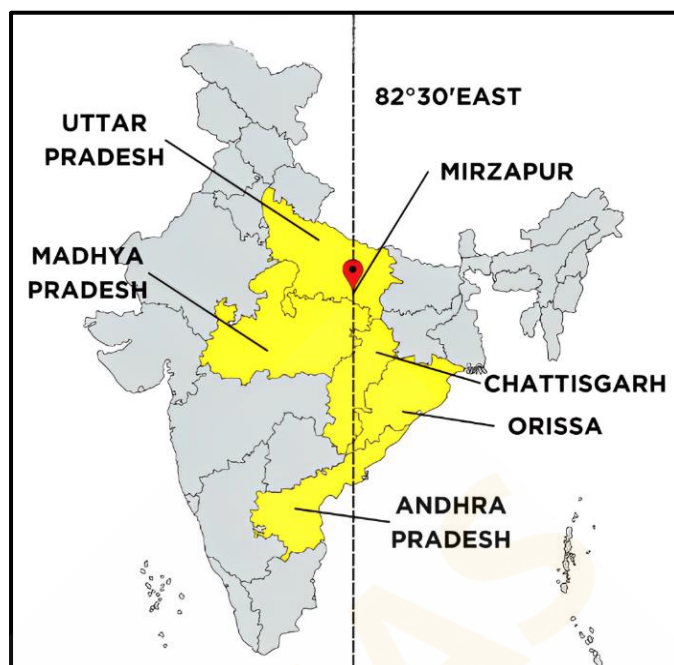
Peninsular Shape and Maritime Influence:

India's geographical shape significantly influences its maritime boundaries. The northern part of India is broad, while the **southern part tapers towards the Indian Ocean**, effectively dividing the northern Indian Ocean into two distinct parts: the **Arabian Sea to the west** and the **Bay of Bengal to the east**. The total length of India's coastline, including the island groups, is approximately **7,516.6 kilometers**. The **Palk Strait**, a narrow body of water, separates the Indian mainland from Sri Lanka. Structurally, Sri Lanka is considered an extension of the Indian peninsular block.

Time Zone and Standard Time:

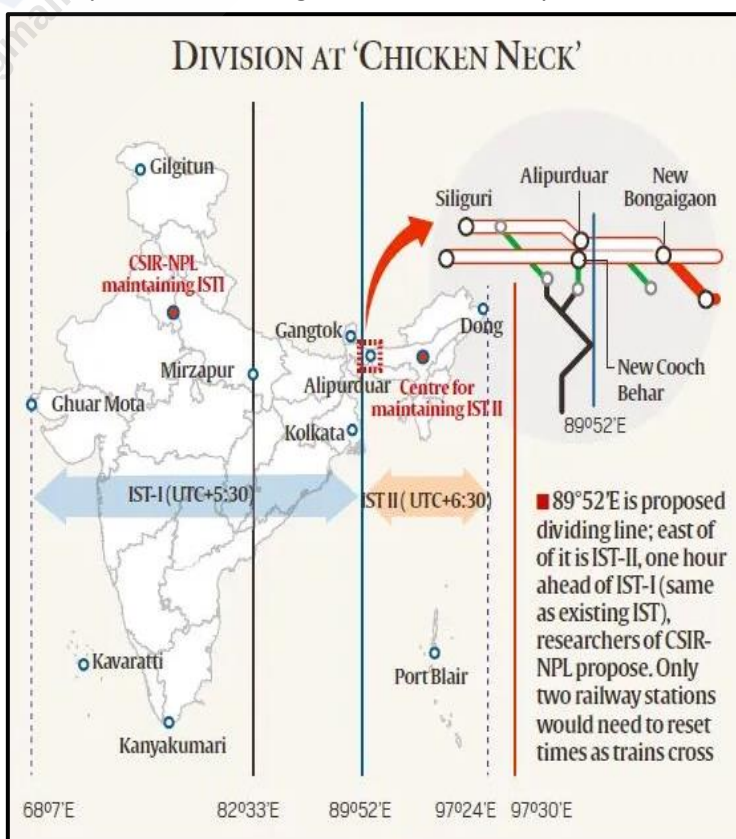
Due to India's **longitudinal extent of nearly 29°** , there is a **real-time difference of about two hours** between its eastern and western extremities. While the eastern extremity experiences sunrise, the western extremity would still have nearly two hours before the day begins. To manage this time difference, India, like other countries, follows a single standard time based on a central meridian.

The standard meridian of India is $82^\circ 30'E$, which has been chosen as it lies approximately in the middle of the country's longitudinal extent. This ensures uniform time across the nation. India's Standard Meridian (IST) passes through **5 states in India - Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh**.



Need for Two Time Zones:

- **Large Geographical Extent** - India is the second-largest country geographically, after China, that operates on a single time zone. The country spans from **$97^\circ 25'E$ in Arunachal Pradesh to $68^\circ 7'E$ in Gujarat**, covering nearly **30 degrees of longitude**, which is sufficient to justify the implementation of two time zones.
- **Early Sunrise and Sunset** - The call for two time zones has gained traction due to the early sunrise and sunset experienced in northeastern India (NER) and the Andaman and Nicobar Islands (ANI). In these regions, the **sun rises and sets much earlier compared to the rest of the country**. The NER has long raised concerns about the impact of a single time zone on daily life and economic activities, as valuable daylight hours are lost. In summer, the sun can rise as early as 4 am, but government offices do not open until 10 am, resulting in inefficiencies.
- **Sync with Circadian Rhythm** - Many Indians are living and working in a time zone that does not align with their natural diurnal cycles. **Productivity and efficiency** are closely tied to the biological clock, which is synchronized with the light-dark cycle. From a circadian rhythm perspective, the current Indian Standard Time (IST) suits Kanyakumari, Kavaratti, and Ghuar Mota well, is manageable for places like Alipurduar, Kolkata, Gangtok, Mirzapur, and Gilgitum, but is highly unsuitable for locations like Dong and Port Blair.
- **CSIR's Suggestions** - In **2006**, the **Planning Commission recommended** splitting the country into two time zones. The **CSIR-National Physical Laboratory (NPL) has endorsed** this long-standing demand, proposing two ISTs separated by a one-hour difference:
 - **IST-I** for most of India, covering regions between **$68^\circ 7'E$ and $89^\circ 52'E$** .
 - **IST-II** for the Northeastern region, covering areas between **$89^\circ 52'E$ and $97^\circ 25'E$** .



Advantages of Having Two Time Zones:

- **Efficiency** - Adopting two time zones could **enhance workforce efficiency and optimize energy consumption**. People would be better able to work and plan according to their natural cycles. For instance, tea gardens in Assam have long operated on an informal time zone known as "**Chaibagaan Time**," which is set one hour ahead of IST.
- **Environmental Protection** - Advancing IST by just half an hour could **save 2.7 billion units of electricity annually** by utilizing the wasted daylight hours. This conservation is crucial for India's economy, which faces a significant power deficit. According to CSIR-NPL, India could **save ₹1,000 crore per year** by conserving electricity through the implementation of two time zones. Reducing energy consumption would also significantly lower India's carbon footprint, supporting its commitment to combating climate change.
- **Societal Benefits** - Two time zones could help **achieve various social policy objectives, such as reducing road accidents and improving women's safety**. Research indicates that people in countries with multiple time zones can better focus on their strengths and weaknesses, leading to improved outcomes. Adjusting the time zone also allowed employees to **develop healthier eating, sleeping, and working habits, yielding positive results**.

Disadvantages of Having Two Time Zones:

- **Timing Mismatch** - The introduction of two time zones could lead to **mismatches in office timings, varying working hours for banks, and potentially more frequent railway accidents**. Implementing two time zones would require careful synchronization of railway traffic to avoid confusion. It could also disrupt travel schedules, as most people travel by road, not air, unlike in the West where adjusting clocks suffices.
- **Chaos at Border of Time Zones** - Given India's significant **illiteracy levels and low awareness**, dividing the country into two time zones could create chaos at the borders of these zones. People would need to reset their clocks each time they cross a time zone boundary. Additionally, marking the dividing line between the two zones would pose a challenge.
- **Prone to Increase Divide** - Implementing two time zones could have **adverse political consequences, potentially exacerbating divisions in India**, which is already divided along lines of religion, caste, race, and language. A separate time zone for the NER could foster a sense of separation from the rest of the country, possibly fueling secessionist sentiments.
- **Infrastructural Burden** - Establishing two time zones would necessitate a second laboratory for CSIR-NPL in the new time zone. This would require setting up a '**Primary Time Ensemble-II**,' traceable to the **Coordinated Universal Time (UTC)** at the International Bureau of Weights and Measures (BIPM) in France.

Chaibagaan Time

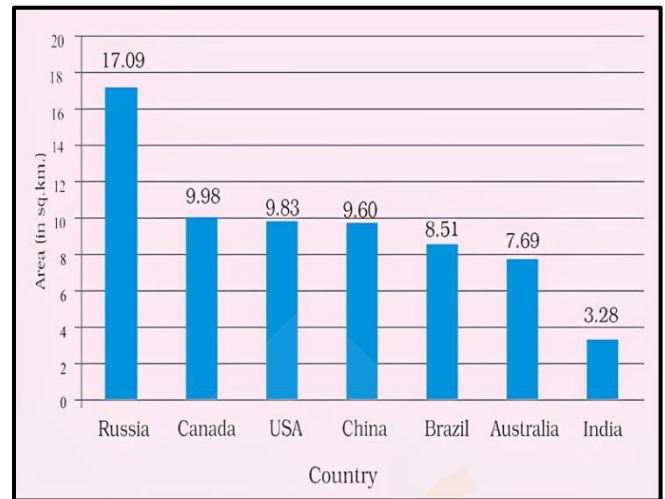
*150 years ago, British colonialists introduced "Chaibagaan time" or "Bagaan time", a time schedule observed by tea planters, which was **one hour ahead of IST**. This was done **to improve productivity by optimizing the usage of daytime**. After Independence, Assam, along with the rest of India, has been following IST for the past 66 years. The administration of the Indian state of Assam now wants to change its time zone back to Chaibagaan time to conserve energy and improve productivity. Indian government didn't accept to such a proposal.*

Size of India:

India stands as the **seventh-largest country** in the world by land area, spanning an extensive **3.28 million square kilometres**. The nation is marked by a significant land boundary of **15,200 km** and a coastal boundary, including its islands, that stretches to 7,516 km. Despite occupying a mere **2.4% of the world's total land area**, India's size and geographical diversity are remarkable.

On the map of India, prominent natural features like the **youthful fold mountains** dominate the landscape, particularly in the **north, northwest, and northeast**, forming **natural boundaries**.

India's coastline extends for 7,516.6 kilometers, including the **mainland coastline of 6,100 kilometers** and the **coastline of Indian islands**, which measures **1,197 kilometers**. This extensive coastal stretch runs through **13 states and union territories of the country**. The formation of India's coastline is linked to the tectonic activities associated with the faulting of Gondwanaland during the Cretaceous period. This geological event, driven by the process known as Continental Drift, resulted in the creation of a straight and regular coastline. However, due to its geological origins, India's coast has a **limited number of naturally favourable harbours**. The drifting and separation of landmasses following the breakup of Gondwanaland during the Cretaceous or early Tertiary periods also played a crucial role in shaping the Indian subcontinent's coastal features, giving rise to the Bay of Bengal and the Arabian Sea.



Significance of India's Location:

India's strategic geographical location has played a pivotal role in shaping its **diplomatic and economic relationships** with other nations. As a subcontinent, India shares **land borders with multiple countries** to the north and south, enhancing its connectivity with diverse regions. The southern half of India extends into the vast Indian Ocean, a factor of significant strategic importance for international trade. The country's **extensive coastline and well-established maritime channels** have facilitated active interaction with the global community. **The naming of the Indian Ocean after India underscores the country's central position within this vital water body.**

Historically, before the comprehensive exploration of sea routes, mountain passes served as crucial pathways for traders and travellers to enter and exit India, highlighting the enduring significance of India's geographical location in facilitating cross-border movement and trade. **India's central positioning in Asia has been a catalyst for fostering trade and promoting its active participation in global affairs.** This advantageous geographic placement has been instrumental in enhancing India's economic and diplomatic engagement with the broader international community.

India and its Neighbours:

Most of India's states have either international borders or coastlines, making them critical from a border management perspective. States like **Madhya Pradesh, Chhattisgarh, Jharkhand, Delhi, Haryana, and Telangana** do not share **international borders or coastlines**. In contrast, other states have one or more international borders or coastlines, categorizing them as frontline states in terms of border security. The country has a land border stretching over **15,106.7 km**, running through **92 districts** across **17 states**.

India shares its **longest land border** with **Bangladesh**, extending over **4,096.7 km**, while its **shortest border**, measuring **106 km**, is with **Afghanistan**.

Name of the country	Length of the border (in Km)
Bangladesh	4,096.7
China	3,488
Pakistan	3,323
Nepal	1,751
Myanmar	1,643
Bhutan	699
Afghanistan	106
Total	15,106.7

Border with China:

India shares its **second-longest border with China**, covering a distance of **3,488 km** running along **Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh** and **Union Territory of Ladakh**. The Sino-Indian border is commonly divided into **three sectors** - the **Western sector**, the **Middle sector**, and the **Eastern sector**.

The Western Sector:

- India and China dispute the **Aksai Chin** region, with India claiming it as part of Ladakh (erstwhile Jammu and Kashmir) and China considering it part of Xinjiang. This dispute originated from the ambiguity in the **British demarcation of the border**, with the **Johnson Line of 1865** placing Aksai Chin under Indian control and the **McDonald Line of 1893** placing it under **Chinese control**. Currently, the **Line of Actual Control (LAC)** separates Indian-administered Ladakh from Aksai Chin, aligning with China's claim line.



The Middle Sector:

- The Middle sector sees **fewer disagreements**, with the boundary running along the watershed between **Himachal Pradesh** and **Uttarakhand** on the Indian side and Tibet on the Chinese side.

The Eastern Sector:

- The Eastern sector dispute centers on the **McMahon Line**, with China rejecting it as an illegal boundary. The dispute is rooted in the **1914 Shimla Convention**, where the Tibetan representatives, according to China, lacked the authority to sign the agreement delineating the McMahon Line.

Border with Nepal:

India shares a **1,751 km** long border with Nepal, touching the Indian states of **Uttarakhand, Uttar Pradesh, Bihar, West Bengal, and Sikkim**. India and Nepal share an **open border, allowing unrestricted movement of goods and people between the two countries**. The boundary largely runs along the foothills of the **Shiwalik Range**.

Disputed Territories:

- **Kalapani** - Kalapani, a valley administered by India as part of Uttarakhand, is situated on the route to **Kailash Mansarovar**. The dispute arises from differing interpretations of the source of the **Kali River**, which demarcates the boundary between India and Nepal as per the **1816 Treaty of Sugauli**.
- **Susta** - Susta, located on the bank of the **Gandak River**, is another disputed area between India and Nepal. The changing course of the Gandak River has led to the shifting of the Susta region, creating a boundary dispute.

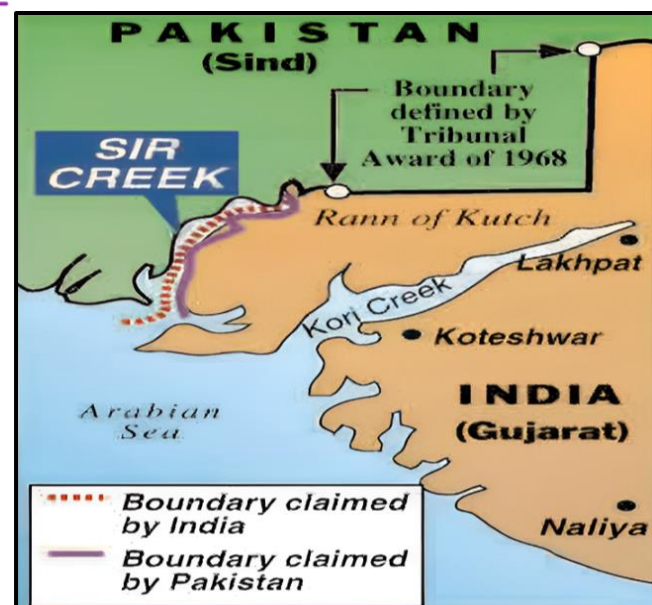


Border with Pakistan:

The Indo-Pakistan boundary, extending **3,323 km** running along **Gujarat, Rajasthan, Punjab, Union Territory of Jammu & Kashmir** and **Union Territory of Ladakh**. Resulted from the partition of India in 1947 under the **Radcliffe Award**, chaired by **Sir Cyril Radcliffe**.

Disputed Territories:

- **Jammu and Kashmir** - Pakistan occupies approximately 78,000 square kilometers of Indian territory in Jammu and Kashmir. Additionally, under the **1963 Sino-Pakistan Boundary Agreement**, Pakistan illegally ceded 5,180 square kilometers of territory in Pakistan-occupied Kashmir to China.
- **Siachen Glacier** - Located in the eastern **Karakoram range** of the Himalayas, the Siachen Glacier has been under **Indian administration since 1984**. The glacier is a highly strategic area, with ongoing military presence from both India and Pakistan.
- **Saltoro Ridge** - The Saltoro Ridge is a subrange of the **Karakoram**, located southwest of the Siachen Glacier. India controls the main peaks and passes, with Pakistani forces occupying the glacial valleys to the west.
- **Sir Creek** - Sir Creek is a **96 km long disputed estuary in the Rann of Kutch**. India and Pakistan differ in their interpretations of the boundary line, with India claiming the centerline and Pakistan claiming the eastern shore as the boundary.



Border with Bangladesh:

India shares its **longest border, 4,096.7 kilometers**, with Bangladesh. This boundary, established under the **Radcliffe Award**, resulted from the partition of the Bengal province.

Border with Myanmar:

The India-Myanmar boundary, spanning **1,643 kilometers**, runs roughly along the watershed between the **Brahmaputra and Ayeyarwady (Irrawaddy) rivers**. The boundary passes through thickly forested regions, including the Mizo Hills, Manipur, and Nagaland on the Indian side, and the Chin Hills, Naga Hills, and Kachin State on the Myanmar side.



Border with Sri Lanka:

India and Sri Lanka are separated by the narrow **Palk Strait**. The closest points are **Dhanushkodi** on the Tamil Nadu coast in India and **Talaimanar** in the **Jaffna peninsula** in Sri Lanka, connected by a group of islets forming **Adam's Bridge**. While the Indo-Sri Lanka boundary is generally peaceful, there has been tension over the ownership of **Kachchatheevu Island**, which India ceded to Sri Lanka in 1974.



===== XXXXX =====

Geological Structure

Introduction:

The geological structure of a country plays a crucial role in understanding the **types and characteristics of rocks and slopes, the physical and chemical properties of soils, the availability of minerals, and the surface and underground water resources**. In India, the geological structure has been classified into several rock systems, each reflecting the country's complex and varied geological history.

Geological Structure and Time Scale:

The term "**geological structure**" refers to the arrangement and deposition of rocks in the earth's crust, often resulting from or influenced by earth movements. It also encompasses the morphological features of rocks, such as the **Gondwana structure**. The Geological Time Scale, a chronological framework for dating various geological formations and life according to their origin, evolution, and extinction, was first developed by Giovanni Ardunia in 1760. The Standard Geological Time Scale, recognized internationally, was established during the International Geological Congress in 1881, Italy. **The Indian Geological Time Scale, advocated by T.S. Holland, provides a specific timeline for geological formations in India.**

India's geological history can be analyzed through its physiographic divisions, including **Peninsular region** and **Extra Peninsular region**. Significant events in India's geological history include the **formation of the Earth's crust, the upheaval of the Himalayas** during the Tertiary period, and the **aggradational formation of the Indo-Gangetic plain** during the Pleistocene period, which continues through sedimentation in the floodplains of the rivers and the lower Gangetic plain.

Major Rock Systems in India:

→ Not mentioned in NCERT

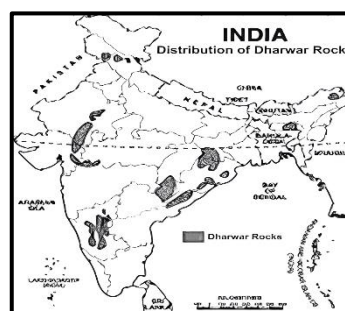
Based on this geological history, the **Geological Survey of India** classifies the country's rock systems into **four major divisions** - Archaean Rock System, Purana Rock System, Dravidian Rock System and Aryan Rock System.

Archaean Rock System (Pre-Cambrian Rocks):

The earliest phase of tectonic evolution marked the **cooling and solidification of the Earth's upper crust** during the Archaean era, which occurred over **2.5 billion years ago**. This system is represented by gneisses and granites, forming the core of the Indian Craton, a block of the Indian subcontinent that was once part of **Gondwanaland**.

The Archaean Rock System includes:

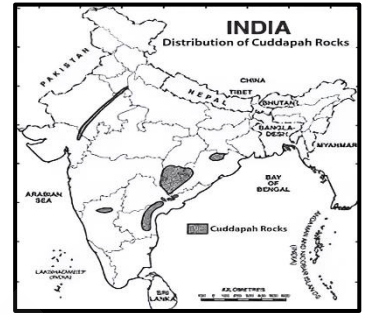
- **Archaean System** - Comprising granites and gneisses, these rocks are the oldest on Earth, formed through the solidification of molten magma. They are also known as the "**Basement Complex**," forming the base for new layers.
- **Dharwar System** - This system represents the **first sedimentary rocks** formed from the metamorphosis of sediments from Archaean gneisses and schists. These rocks are **highly metamorphosed, economically important** due to their mineral content, and found abundantly in **Karnataka's Dharwar district**.



Purana Rock System:

The Purana Rock System, consisting of the **Cuddapah and Vindhyan rock systems**, was formed between **1400-600 million years ago** through the **erosion and deposition of Archean and Dharwar rocks**. These rocks are primarily sedimentary.

- **Cuddapah System** - Named after the **Cuddapah district in Andhra Pradesh**, this system features sedimentary rocks like **sandstone and limestone**, deposited in synclinal folds. These rocks contain ores of iron, manganese, and other minerals.
- **Vindhyan System** - Deriving its name from the **Vindhyan mountains**, this system comprises ancient sedimentary rocks, including **diamond-bearing regions like Panna and Golconda**. Although devoid of metalliferous minerals, the system provides large quantities of durable stones and glass-making sand.



Dravidian Rock System (Palaeozoic):

Formed during the Paleozoic era (**600-300 million years ago**), the Dravidian Rock System primarily consists of sedimentary rocks found in **extra-Peninsular regions** like the Himalayas and Gangetic plain. The system marks the beginning of life on Earth, with rocks containing plenty of fossil evidence. The Dravidian period also saw the formation of **high-quality Carboniferous coal**, although these formations are **not abundant in India**. Key components of the Dravidian system include:

- **Carboniferous Rocks** - Formed around **350 million years ago**, these rocks primarily consist of **limestone, shale, and quartzite**. Notably, Mount Everest is composed of Upper Carboniferous limestone.

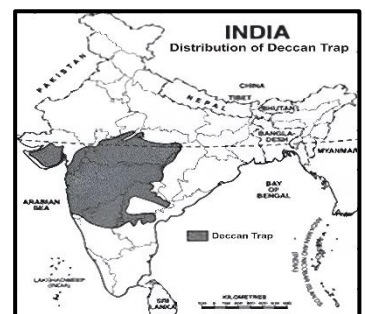
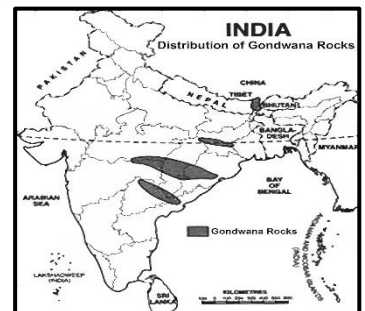
Aryan Rock System:

Beginning in the **Upper Carboniferous period**, the Aryan Rock System extends from the Upper Carboniferous to the Holocene period.

This Aryan system is classified into several groups:

→ Deccan Plateau
→ Andhra Pradesh
→ Kutch
→ Himalayan Region

- **Gondwana System** - Named after the **Gonds, an indigenous people of Telangana and Andhra Pradesh**, this system consists of deposits laid down in synclinal troughs on the ancient plateau surface since the **Permian period (250 million years ago)**. The Gondwana rocks contain nearly **98% of India's coal reserves**, along with other valuable minerals like iron ore and copper.
- **Jurassic System** - This system features marine transgressions that led to thick shallow water deposits in regions like **Rajasthan and Kachchh**, with coral limestone, sandstone, and shales.
- **Deccan Trap** - Resulting from volcanic outbursts over **Peninsular India** between the **Cretaceous and Eocene periods**, the Deccan Trap covers a vast area with basaltic lava flows. Weathering of these rocks has given rise to **black cotton soil** known as 'regur.'
- **Tertiary System** - The Tertiary period, spanning from the Eocene to the **Pliocene (60-7 million years ago)**, is significant in India's geological history as it saw the upliftment of the Tethys geosyncline, leading to the formation of the Himalayas. The Tertiary Succession is found in regions like the **Bengal and Ganges delta**, and the **outer Himalayan regions**.



A Diverse and Unique History:

India's geological history is marked by its unique and diverse characteristics, with different regions containing rocks from various geologic periods. The Indian Craton was once a part of the supercontinent Pangaea. Following the **breakup** of the **Gondwanaland craton** around **225 million years ago** and its **movement toward the Eurasian craton**, the **Himalayan Mountain range** began to form **approximately 65 million years ago**.

In the aftermath of the **Outer Himalayas' formation**, **widespread alluvial plains** started developing during the **Upper Pliocene and Pleistocene epochs**, leading to the creation of the Indo-Ganga-Brahmaputra plain. As a result, different regions of India feature rocks representing nearly every type of geological structure from various geologic periods.

The stratigraphy of India can be categorized into several divisions, including the **Archean System**, **Dharwar System**, **Cuddapah System**, **Vindhyan System**, **Dravidian System**, and **Aryan System** (which includes the Gondwana, Jurassic, Deccan Trap, Tertiary, and Quaternary rocks).

India's geological strata span from the **oldest rocks to the newest alluvial** formations. The ancient rocks of the Archaean period are predominantly found in peninsular India. Sedimentary rocks are prevalent in areas formed by sediment deposition from the Indo-Gangetic plains. Additionally, various sedimentary and metamorphic rocks can be traced in the Great Folded Mountainous region.

